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Class 10 Biology

Chapter 2 **Notes+Short Questions** **+Long Questions** **NBF book**

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Chapter no 2

SLO QUESTION AND ANSWER

Q1. Why do athletes train at high altitudes?

- **Answer:**
 - High places have less oxygen in the air.
 - The body makes more red blood cells to cope.
 - This helps the blood carry oxygen much better.
 - It gives athletes more energy when they return.
-

Q2. How does lacking a nucleus benefit mature red blood cells?

- **Answer:**
 - It gives the cells a flat, disc-like shape.
 - This shape provides more surface area for oxygen.
 - It leaves more space inside to pack hemoglobin.
 - It helps cells twist easily through small tubes.
-

Q3. Why does blood serum fail to form clots?

- **Answer:**
- Serum is blood liquid without clotting tools.
- It does not have the protein called fibrinogen.
- Because of this, it cannot make a clot mesh.
- The liquid stays watery and cannot become solid.

Q4. What happens if a patient has a low platelet count?

- **Answer:**
 - The body cannot stop bleeding normally.
 - Platelets cannot group together to plug a wound.
 - Even small cuts will bleed for a long time.
 - The person will easily get internal bruises.
-

Q5. Why are white blood cells transparent?

- **Answer:**
 - They do not have any colored gas-carrying pigments.
 - They completely lack the red protein called hemoglobin.
 - Doctors must use colored dyes to see them.
-

Q6. Why is the left ventricular wall exceptionally thick?

- **Answer:**
- It has to push blood to the whole body.
- Pushing blood that far takes a lot of force.

- o It needs strong muscle to create high pressure.
 - o The right side only pushes blood to the nearby lungs.
-

Q7. Why does a heartbeat without nervous input?

- **Answer:**
 - o The heartbeat starts inside the heart muscle itself.
 - o It does not need brain signals to make it beat.
 - o Natural built-in cells control the steady rhythm.
 - o Brain nerves only tell it to go faster or slower.
-

Q8. How do atrioventricular valves prevent blood backflow?

- **Answer:**
 - o The lower heart chambers squeeze blood out tightly.
 - o This pressure pushes hard against the valve doors.
 - o The doors slam shut together instantly.
 - o Blood is blocked from moving backward into the top chambers.
-

Q9. Why do atria have very thin walls?

- **Answer:**
- o They only act as waiting rooms for blood.
- o They push blood a very short distance.

- o The blood goes straight down into the ventricles.
 - o This action requires very little muscle power.
-

Q10. What physically generates the "lub-dub" heartbeat sound?

- **Answer:**
 - o The "lub" sound is the first atrioventricular valves shutting.
 - o This happens when the lower chambers start squeezing.
 - o The "dub" sound is the exit semilunar valves shutting.
 - o This happens when the lower chambers relax.
-

Q11. Why do veins require internal valves while arteries do not?

- **Answer:**
 - o Blood moves slowly and under low pressure in veins.
 - o It often has to travel upward against gravity.
 - o Valves stop the blood from sliding back down.
 - o High pressure in arteries keeps blood moving forward anyway.
-

Q12. How do capillaries optimize nutrient diffusion?

- **Answer:**
- o Their walls are made of just one thin layer.

- o Food and gases have a very short path to cross.
 - o Blood flows very slowly inside these tiny tubes.
 - o This gives materials plenty of time to swap.
-

Q13. Why does the pulmonary artery carry deoxygenated blood?

- **Answer:**
 - o All vessels leaving the heart are named arteries.
 - o This specific tube leaves the heart for the lungs.
 - o It takes used blood there to get fresh oxygen.
 - o Therefore, it holds blood that lacks oxygen.
-

Q14. What makes the hepatic portal vein unique?

- **Answer:**
 - o It connects two different organ networks together.
 - o It takes blood straight from stomach to liver.
 - o It does not go to the main body vein first.
 - o This lets the liver clean food nutrients immediately.
-

Q15. Why arterioles are called resistance vessels?

- **Answer:**

- o They have active muscles inside their walls.
 - o They can squeeze tight or open up wide.
 - o This limits how much blood enters tiny capillaries.
 - o This squeezing directly controls total body blood pressure.
-

Q16. Why is carbon monoxide exposure fatal?

- **Answer:**
 - o It locks onto red cells very strongly.
 - o It takes the spot where oxygen should sit.
 - o It completely blocks oxygen from entering the blood.
 - o Body parts die quickly because they cannot breathe.
-

Q17. How do dietary fats enter circulation?

- **Answer:**
 - o Fats cannot enter normal, tight blood tubes.
 - o They are absorbed by special tubes called lacteals.
 - o They travel through the body's fluid system first.
 - o They join the main bloodstream much later.
-

Q18. How does a moving embolus cause a heart attack?

- **Answer:**

- A loose blood clot floats through the blood.
 - It gets stuck in a narrow heart artery.
 - It blocks blood from reaching the heart muscle.
 - That part of the heart muscle dies quickly.
-

Q19. Why angina is not considered a true disease?

- **Answer:**

- It is a warning sign, not the main illness.
- It shows the heart is briefly short on oxygen.
- It causes pain when the heart works too hard.
- It tells you that heart tubes are getting blocked.

Q20. How does nicotine degrade cardiovascular health?

- **Answer:**

- It makes blood vessels squeeze and become narrow.
 - This instantly forces blood pressure to go up.
 - It makes the heart beat much faster.
 - The heart gets tired from working too hard.
-

CH # 2 CIRCULATORY SYSTEM

LONG QUESTIONS

Q1. Describe the Pulmonary and Systemic Circuits of the Circulatory System.

The human circulatory system is a **double circulatory system**, meaning blood passes through the heart twice during one complete trip around the body. This is divided into two major loops: the pulmonary circuit and the systemic circuit.

1. The Pulmonary Circuit

The word "pulmonary" relates to the lungs. The main job of this circuit is to pick up oxygen and drop off carbon dioxide.

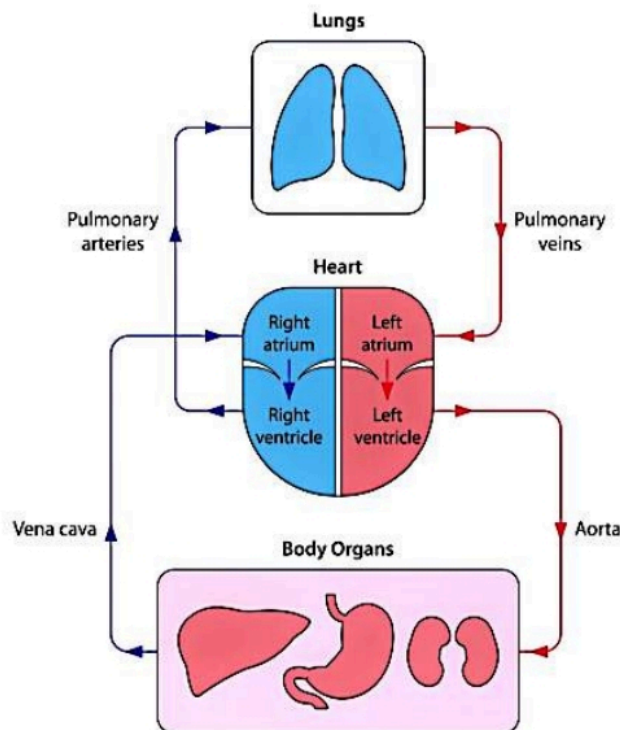
- **The Pathway:** It begins at the **right ventricle**, which pumps oxygen-poor (deoxygenated) blood into the **pulmonary artery**. This artery splits to carry the blood to both **lungs**.
- **The Exchange:** Inside the lungs, blood releases carbon dioxide gas (a waste product) and absorbs fresh oxygen from the air you breathe in.
- **The Return:** The newly oxygen-rich (oxygenated) blood travels back to the heart via the **pulmonary veins**, emptying into the **left atrium**.

2. The Systemic Circuit

The systemic circuit deals with the rest of your body's systems. Its job is to deliver oxygen and nutrients to every cell and collect waste.

- **The Pathway:** It starts at the highly muscular **left ventricle**, which forces oxygenated blood out through the body's largest artery, the **aorta**.
- **The Delivery:** Branching arteries carry this blood to the head, arms, digestive organs, kidneys, and legs. Capillaries distribute nutrients and oxygen to tissue cells and pick up metabolic waste.
- **The Return:** The now-deoxygenated blood collects into large veins called the **vena cava** (superior and inferior) and returns safely to the **right atrium** of the heart to begin the process all over again.

CIRCULATORY SYSTEM DIAGRAM



Q2. How Materials Are Transported Through the Blood?

Blood acts as the body's ultimate highway system, moving diverse chemical materials simultaneously using different transport mechanisms:

- **Oxygen Transport (O_2):** Roughly 98.5% of oxygen does not dissolve well in water, so it hitches a ride on **hemoglobin**, an iron-rich protein packed inside red blood cells. When oxygen binds to it, it forms a temporary compound called **oxyhemoglobin**, which safely dissociates (splits) when it reaches oxygen-starved tissues.
- **Carbon Dioxide Transport (CO_2):** Carbon dioxide is transported in three ways:
 1. The majority (around 70%) is converted chemically into **bicarbonate ions** (HCO_3^-) dissolved right in the liquid plasma.
 2. About 23% binds directly to hemoglobin proteins, forming **carbaminohemoglobin**.
 3. Around 7% simply dissolves directly into the plasma water.
- **Nutrients and Wastes:** Digested food monomers (like glucose, amino acids, and fatty acids), salts, and vitamins are completely water-soluble and travel dissolved natively in the plasma liquid. Chemical wastes, like **urea** from the liver, dissolve in the plasma too, traveling smoothly to the kidneys to be filtered out into urine.
- **Hormones:** Chemical messengers secreted by endocrine glands (like adrenaline or insulin) are released directly into the blood plasma, which carries them across the body to trigger specific target organs.

Q3. What are the organs connected to the Blood Circulatory System

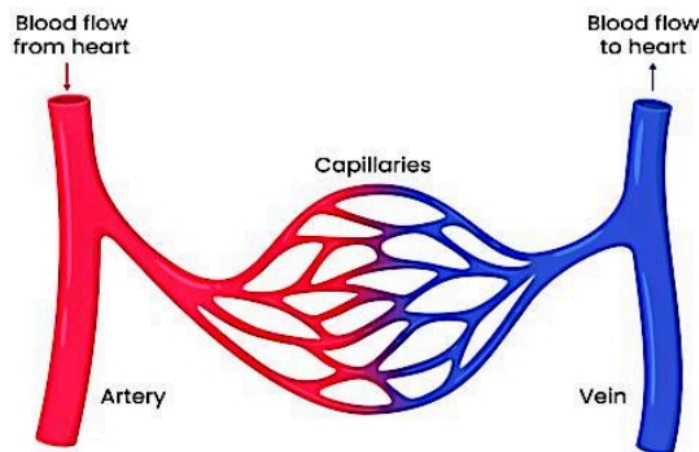
The cardiovascular loop interacts with nearly every major organ system, each performing a unique exchange function:

Connected Organ	Main Role in the Circulatory System
The Lungs	Primary site for gas exchange . They add fresh oxygen directly into the pulmonary blood circuit and remove volatile carbon dioxide waste.
The Small Intestine	The primary site for nutrient absorption . Blood vessels lining the intestinal walls absorb glucose, amino acids, and vitamins, distributing them to body cells.
The Liver	Acts as a chemical processing plant . It receives nutrient-dense blood directly from the intestines via the hepatic portal vein to filter toxins, store vitamins, and regulate blood glucose levels.
The Kidneys	Act as the filtration system . They filter metabolic waste products (like urea and excess salts) out of circulating blood to maintain blood volume and proper pH.
The Bone Marrow	The production factory . Spongy bone marrow contains stem cells that continuously manufacture new red blood cells, white blood cells, and platelets.

Q 4 What is the role of blood vessels?

The primary role of blood vessels is to act as a closed network of tubes that conducts blood safely throughout the body. Instead of isolated plumbing pipes, they form a continuous, tapering loop that connects the heart to the tissues and back again.

Here is how materials flow straight through all five vessel types :



1. Arteries

- **Role:** Carry high-pressure, oxygenated blood away from the heart (except the pulmonary artery).
- **Structure:** They have thick, muscular, and highly elastic walls to handle the forceful surges of blood pumped directly by the ventricles.

2. Arterioles

- **Role:** Act as the "control valves" regulating blood flow.
- **Structure:** As large arteries travel deeper into organs, they branch into these much smaller vessels. Their muscular walls constrict or dilate to adjust blood pressure and dictate exactly how much blood reaches specific tissues.

3. Capillaries

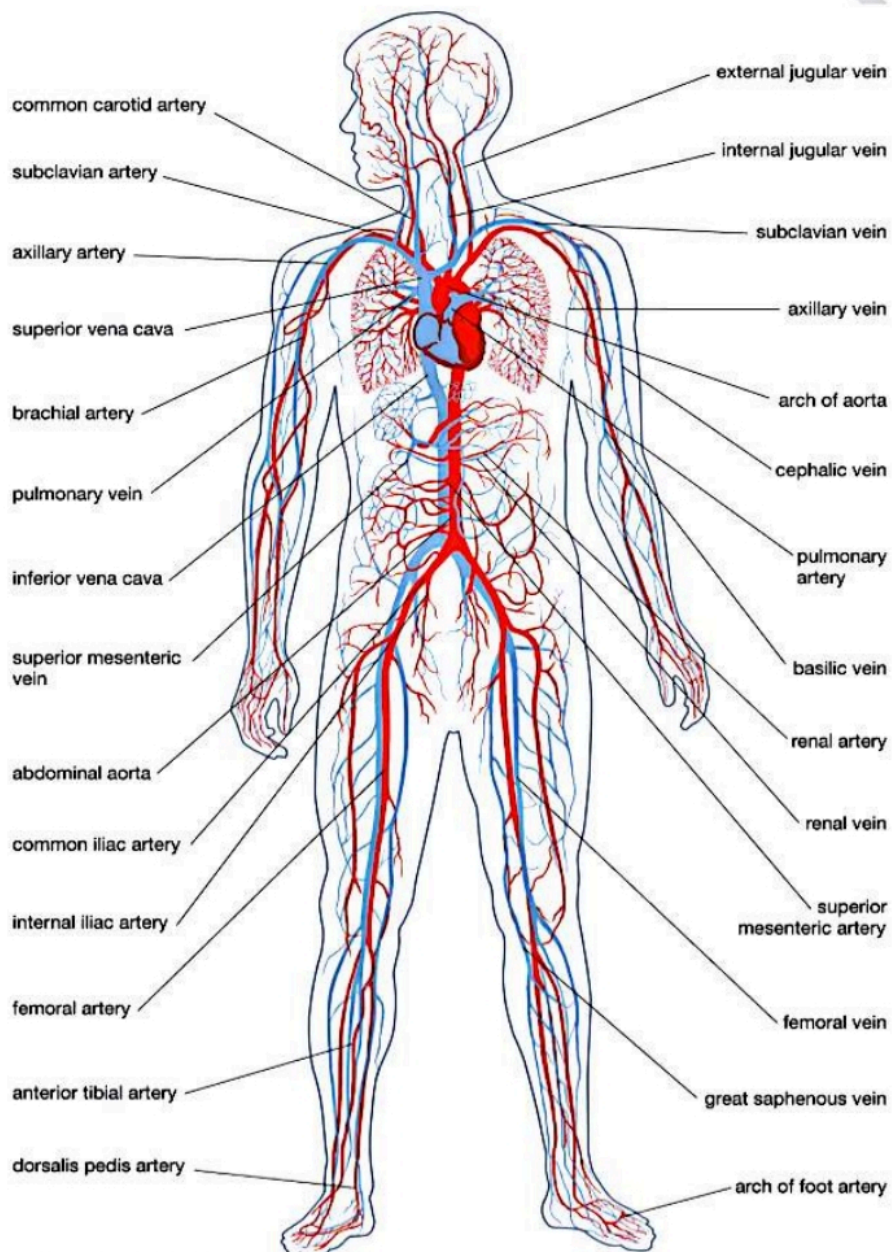
- **Role:** The unique site of **material exchange**.
- **Structure:** Arterioles branch into these microscopic networks. Their walls are incredibly thin—**only one cell thick**. This ultra-thin design allows oxygen, glucose, and nutrients to easily pass out into body cells, while carbon dioxide and cellular wastes diffuse into the blood.

4. Venules

- **Role:** Collect waste-heavy blood from the capillary beds.
- **Structure:** These are tiny vessels formed when capillaries begin merging back together. They serve as the initial collection pipes for deoxygenated blood on its return journey.

5. Veins

- **Role:** Carry low-pressure, deoxygenated blood back **To** the heart (except pulmonary veins).
- **Structure:** Venules merge into larger veins. Because blood pressure here is very low, veins have thin walls but contain vital **one-way internal valves** to fight gravity and prevent blood from flowing backward.



Q5. What are the Components of Blood?

If you were to take a sample of blood and spin it rapidly in a machine called a centrifuge, it would separate into two distinct layers based on weight: a liquid layer at the top and a dense, cellular layer at the bottom.

From this, we learn that blood is a fluid connective tissue made of two main components: **Blood Plasma** and **Formed Elements (Blood Cells)**.

1. Blood Plasma (The Liquid Component)

Plasma makes up about **55%** of your total blood volume. It is a straw-colored, slightly sticky liquid that acts as the liquid highway of the circulatory system.

- **Water (90-92%):** The vast majority of plasma is water. It acts as a solvent and helps maintain body temperature by absorbing and distributing heat.
- **Plasma Proteins (7-8%):** These are vital proteins manufactured mostly by the liver:
 - *Fibrinogen*: Essential for blood clotting.
 - *Albumin*: Maintains osmotic pressure, keeping fluid from leaking out of blood vessels into surrounding tissues.
 - *Globulins*: Act as antibodies to protect the body against infections.
- **Dissolved Solutes (1-2%):** This includes nutrients (glucose, amino acids, lipids), metabolic wastes (urea, uric acid), gases (carbon dioxide), and electrolytes (sodium, potassium, and calcium ions).

2. Formed Elements (The Cellular Component)

The remaining **45%** of blood consists of solid cellular structures suspended within the plasma. These are produced in the red bone marrow and include:

- Red Blood Cells (Erythrocytes)
- White Blood Cells (Leukocytes)
- Platelets (Thrombocytes)

Q6. Describe the Cell Types Found in Blood

There are three distinct types of cells (and cell fragments) floating inside our blood plasma. Each one has a completely different shape and structure, specifically adapted to perform a unique job to keep you alive.

1. Red Blood Cells (RBCs) or Erythrocytes

Red blood cells are by far the most numerous cells in your body. A single drop of human blood contains about 5 million of them!

- **Structure:** They are shaped like **biconcave discs**—which means they look like a donut with a thin, dimpled center instead of a hole. This unique dimple gives them a massive surface area to absorb and release gases quickly. Curiously, as an RBC matures, it **loses its nucleus** and mitochondria. This unique adaptation leaves maximum room to

pack in hemoglobin and ensures the cell doesn't consume the oxygen it is supposed to be delivering.

- **Function:** Their main job is **oxygen transport**. They are packed with an iron-rich protein called **hemoglobin**, which chemically binds to oxygen in your lungs to form *oxyhemoglobin*. As blood flows through the body, hemoglobin drops off this oxygen to needy tissues.
- **Lifespan:** Because they lack a nucleus to repair themselves, they only live for about 120 days before being recycled by your liver and spleen.

2. White Blood Cells (WBCs) or Leukocytes

White blood cells are the frontline soldiers of your immune system. They are much larger than RBCs, far fewer in number, and contain a distinct, well-defined nucleus.

- **Special Ability:** Unlike red blood cells, which must stay inside the blood vessels, WBCs can actively change their shape like an amoeba. This allows them to squeeze through tiny pores in capillary walls to attack germs directly inside infected body tissues. This process of squeezing through blood vessel walls is called **diapedesis**.

To do their jobs effectively, the white blood cell army is divided into two main categories based on whether they have visible chemical spots (granules) in their cytoplasm when viewed under a microscope: **Granulocytes** and **Agranulocytes**. Altogether, there are five distinct subtypes.

Category A: Granulocytes (Spotted Look)

These cells contain tiny, chemical-filled sacs (granules) in their cytoplasm that look like small spots. Their nuclei are uniquely split into multiple connected segments or "lobes."

- **Neutrophils (The First Responders):** These are the most abundant white blood cells. They have a distinctive **multi-lobed nucleus** (looking like 3 to 5 connected clumps). They act as the body's frontline patrol, quickly rushing to an injury site to swallow up and digest invading bacteria through a process called **phagocytosis**.
- **Eosinophils (The Parasite Fighters):** These cells typically feature a **two-lobed (bilobed) nucleus** that looks a bit like a pair of headphones. Their internal granules release powerful enzymes designed to destroy larger parasites, like microscopic worms, and they also help control allergic reactions.
- **Basophils (The Allergy Triggers):** This is the rarest type of WBC. Their cytoplasm is so densely packed with dark purple spots that it often completely hides their **S-shaped nucleus**. They release a chemical called **histamine**, which causes inflammation and swelling, helping to bring more blood flow—and more fighter cells—to an injured area.

Category B: Agranulocytes (Smooth Look)

These cells have a clear, smooth cytoplasm without visible chemical grains, and their nuclei are solid rather than split into separate lobes.

- **Lymphocytes (The Intelligence Unit):** These are slightly smaller cells dominated by a massive, dark, **perfectly round nucleus** that takes up nearly the entire space inside the cell. They do not eat germs directly; instead, they act as the strategic brains of your immune system. They recognize specific germs you've encountered before and

manufacture specialized chemical weapons called **antibodies** to target and neutralize specific viruses and toxins.

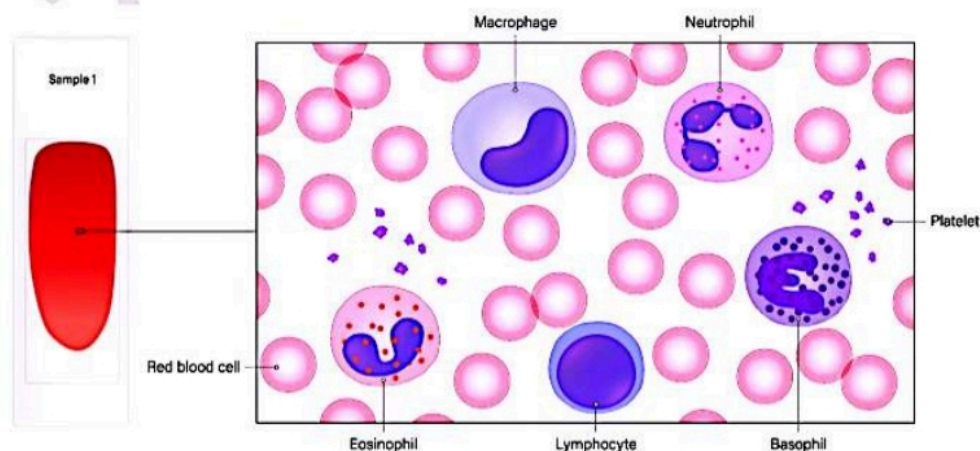
- **Monocytes (The Heavy Clean-up Crew):** These are the largest of all the white blood cells and feature a distinctive **kidney-bean or horseshoe-shaped nucleus**. When there is a major or long-term infection, monocytes travel out of the blood into the tissues, grow even larger, and act as giant scavengers that vacuum up dead cells, cellular debris, and lingering pathogens.

Main Function of WBCs: To defend the body against infections, foreign invaders, and diseases by destroying pathogens, producing antibodies, and clearing away cellular waste.

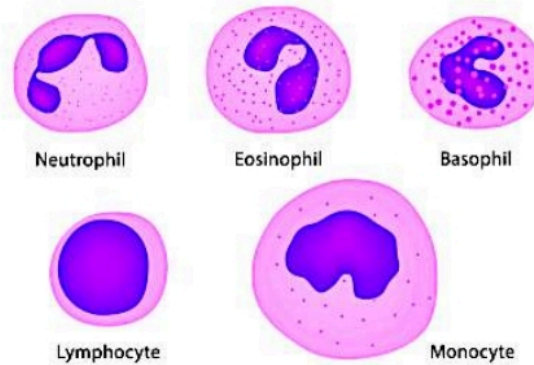
3. Platelets or Thrombocytes

Platelets are not actually whole, complete cells. Instead, they are tiny, membrane-bound cellular fragments.

- **Structure:** They are minute, irregular, disc-shaped structures that completely lack a nucleus. They are formed when giant cells in your bone marrow (called megakaryocytes) shatter into thousands of tiny, microscopic pieces.
- **Function:** They are responsible for **blood clotting (hemostasis)**. If you cut your finger, platelets instantly rush to the injury site, stick together to form a physical plug, and release chemical signals. These signals convert a soluble protein in your plasma into sticky, web-like threads called *fibrin*. This molecular web traps passing blood cells to form a solid scab, stopping you from bleeding and preventing germs from entering.



White Blood Cells



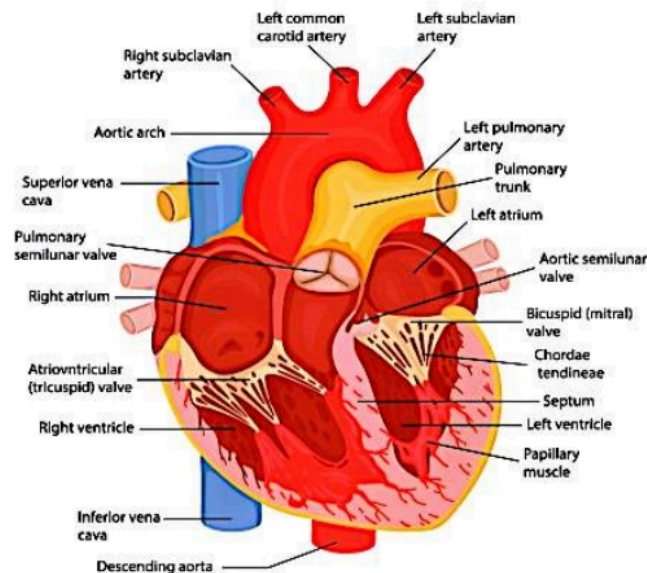
Q7. Draw and describe the structure of the Human Heart

The human heart is a hollow, muscular organ roughly the size of a closed fist, located inside the thoracic cavity just behind the sternum and tilted slightly to the left.

Core Structural Features

- **The Layers:** The heart wall is composed primarily of a specialized muscle tissue called **myocardium**, enclosed externally by a tough protective sac called the **pericardium** and lined internally by the smooth **endocardium**.
- **The Four Chambers:** The heart is split into upper and lower sections:
 - **Right and Left Atria (Singular: Atrium):** The two thin-walled upper chambers that receive returning blood.
 - **Right and Left Ventricles:** The two thick-walled lower chambers that forcefully pump blood out of the heart. The *left ventricle* wall is three times thicker than the right because it must generate enough force to push blood across the entire body.
- **The Septum:** A solid muscular wall called the **interventricular septum** physically separates the left and right sides of the heart, preventing oxygen-rich and oxygen-poor blood from ever mixing.
- **Internal Heart Valves:** To guarantee blood moves in a single forward direction, the heart uses four mechanical valves:
 - **Tricuspid Valve:** Located between the right atrium and right ventricle.
 - **Bicuspid (Mitral) Valve:** Located between the left atrium and left ventricle.
 - **Pulmonary Semilunar Valve:** Guarding the exit from the right ventricle into the pulmonary artery.
 - **Aortic Semilunar Valve:** Guarding the exit from the left ventricle into the systemic aorta.

Human heart anatomy

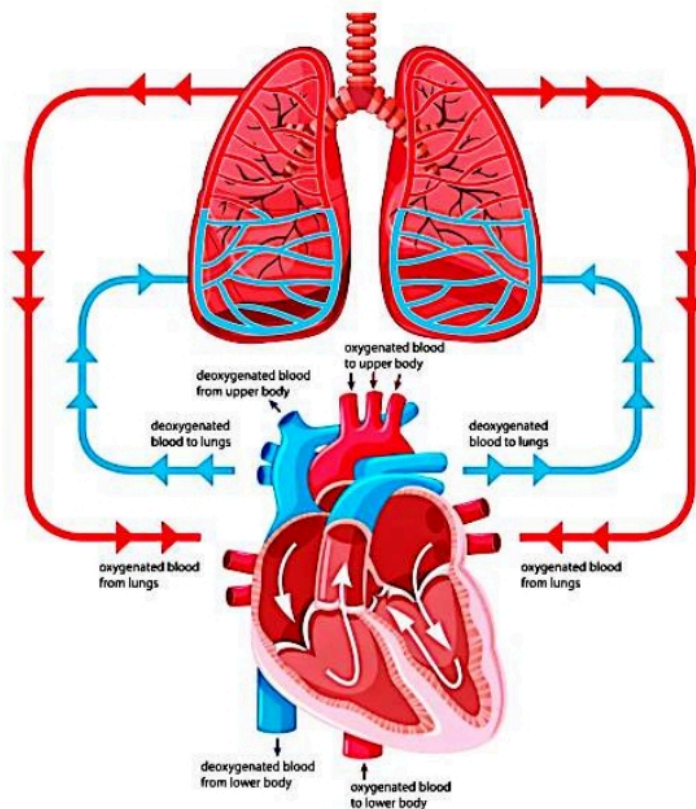


Q8. Explain the Circulation of Blood Through the Human Heart

Blood flows through the heart in a precise, cyclical sequence driven by coordinated muscle contractions:

The Step-by-Step Flow Path

1. **Deoxygenated blood** returning from upper body tissues enters the **Right Atrium** through the **Superior Vena Cava**, while blood from lower body regions enters via the **Inferior Vena Cava**.
2. The Right Atrium contracts, pushing blood down through the open **Tricuspid Valve** into the **Right Ventricle**.
3. The Right Ventricle contracts. The tricuspid valve snaps shut to prevent backflow, forcing blood out through the **Pulmonary Semilunar Valve** into the **Pulmonary Artery**, which carries it directly to the lungs.
4. In the lungs, the blood drops off carbon dioxide and absorbs fresh oxygen.
5. This fresh, **oxygenated blood** travels through the **Pulmonary Veins** and empties into the **Left Atrium**.
6. The Left Atrium contracts, driving blood through the open **Bicuspid (Mitral) Valve** into the **Left Ventricle**.
7. The thick, powerful **Left Ventricle** contracts. The bicuspid valve slams shut, and blood is forced out through the **Aortic Semilunar Valve** into the **Aorta**, heading off to supply the body's tissues.



Q9. Discuss the following Heart Diseases

When structural components of the cardiovascular network degrade or get blocked, serious cardiovascular disorders can occur:

1. Atherosclerosis

Atherosclerosis is a chronic disease where the inside of arteries becomes narrow and rigid.

- **The Process:** It occurs when circulating low-density cholesterol, fats, and cellular waste deposits gather inside artery walls, forming hard structures called **plaques**.
- **The Result:** This restricts smooth arterial blood flow and elevates blood pressure. If a plaque ruptures, it can form a local blood clot (thrombus), cutting off downstream blood flow completely.

2. Arteriosclerosis

While often confused with atherosclerosis, arteriosclerosis is a broader medical term referring to the generic **hardening and loss of elasticity** in the arteries. As individuals age, calcium deposits can build up within the muscular walls of large arteries, causing them to lose their natural ability to expand and contract with each pulse wave. This stiffness forces the heart to work much harder to move blood.

3. Angina Pectoris

Angina pectoris translates literally to "strangled chest." It is not a standalone disease, but a symptom showing that the heart muscle itself isn't receiving enough oxygen-rich blood.

- **The Cause:** This usually happens when the **coronary arteries** (the heart's personal blood supply lines) are partially narrowed by plaque.
- **The Sensation:** During physical exertion or severe emotional stress, the heart beats faster and demands more oxygen. Because the narrowed arteries cannot supply it, the heart muscle experiences temporary oxygen starvation, causing a crushing, suffocating pain behind the breastbone that can radiate down the left arm. It serves as an early warning sign of a potential heart attack.

Q10. Explain the Harmful Effects of Smoking on Heart Health

Smoking tobacco introduces thousands of toxic chemicals into the bloodstream, directly damaging the structural integrity of the cardiovascular system:

- **Nicotine Action:** Nicotine acts as a powerful stimulant. It triggers an immediate release of adrenaline, which forces the heart to beat faster and constricts peripheral blood vessels. This raises blood pressure, forcing the heart muscle to pump much harder against increased resistance.
- **Carbon Monoxide Toxicity:** Cigarette smoke contains carbon monoxide gas. When inhaled, carbon monoxide binds to blood hemoglobin with an affinity over 200 times stronger than oxygen, forming **carboxyhemoglobin**. This reduces the blood's capacity to transport oxygen, leaving vital tissues and the heart muscle oxygen-starved.
- **Plaque Acceleration:** Chemical compounds within cigarette smoke damage the smooth inner endothelial lining of the arteries. This structural damage invites cholesterol deposits, accelerating the progression of atherosclerosis.
- **Increased Blood Stickiness:** Smoking increases the stickiness of blood platelets and raises circulating levels of fibrinogen. This increases the risk of dangerous internal blood clots forming inside narrowed coronary arteries, which can lead to a heart attack or stroke.

BIOLOGY
CLASS: 10TH

CHAPTER 2 : CIRCULATORY SYSTEM

SHORT ANSWER QUESTIONS :

1. Write the functions of:

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1. **.Pericardium:**
A double-layered membrane that surrounds and protects the heart, reduces friction during heartbeats, and anchors it within the chest cavity.
2. **Left atrium:**
Receives oxygenated blood from the lungs via the pulmonary veins and pumps it into the left ventricle.
3. **Right atrium:**
Receives deoxygenated blood from the body via the superior and inferior vena cava and pumps it into the right ventricle.
4. **Left ventricle:**
Pumps oxygenated blood into the aorta for distribution to the entire body. It has the thickest muscle wall.
5. **Right ventricle:**
Pumps deoxygenated blood into the pulmonary artery towards the lungs for oxygenation.
6. **Bicuspid valve (Mitral valve):**
Located between the left atrium and left ventricle, prevents backflow of blood into the left atrium.
7. **Tricuspid valve:**
Located between the right atrium and right ventricle, it prevents backflow into the right atrium.
8. **Semilunar valve:**
Located at the exits of the ventricles (pulmonary and aortic valves), prevent backflow of blood into the heart after contraction.
9. **Septum:**
A muscular wall that divides the left and right sides of the heart and prevents the mixing of oxygenated and deoxygenated blood.
10. **Aorta:**
The largest artery in the body carries oxygenated blood from the left ventricle to the entire body.
11. **Pulmonary artery:**
Carries deoxygenated blood from the right ventricle to the lungs for oxygenation. Only artery that carries deoxygenated blood.

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12. Pulmonary vein:

Carries oxygenated blood from the lungs to the left atrium. The only vein that carries oxygenated blood

13. Plasma:

The fluid part of blood that transports nutrients, hormones, waste products, and other substances.

14. Red blood cells:

contains haemoglobin and transports oxygen from the lungs to tissues and carbon dioxide back to the lungs.

15. White blood cells:

Fight infections and provide immunity by attacking bacteria, viruses, and other foreign substances.

16. Platelets:

Help in blood clotting to prevent excessive bleeding at injury sites.

2. Differentiate between:

a. Serum and blood:

- **Blood:** A complete fluid containing plasma, red blood cells, white blood cells, and platelets.
- **Serum:** Blood plasma without clotting factors (fibrinogen); obtained after clotting has occurred.

b. Artery and vein:

- **Artery:** Carries blood away from the heart (usually oxygenated except pulmonary artery), thick-walled, high pressure.
- **Vein:** Carries blood to the heart (usually deoxygenated except pulmonary vein), thin-walled, has valves, low pressure.

c. Tricuspid and bicuspid valve:

- **Tricuspid Valve:** Between right atrium and right ventricle, has three cusps.
- **Bicuspid Valve (Mitral):** Between left atrium and left ventricle, has two cusps.

d. Systole and diastole:

- **Systole:** Contraction phase of the heart when blood is pumped out.
- **Diastole:** Relaxation phase when the heart chambers fill with blood.

3. Why is a circulatory system necessary?

The circulatory system is essential because it transports oxygen and nutrients to all the cells in the body. It also removes waste products like carbon dioxide. Without this system, body cells wouldn't get the materials they need to function properly, and harmful substances would build up in the body.

4. Name the three circuits of the circulatory system.

The three main circuits of the circulatory system are:

1. **Systemic circulation:** Carries oxygen-rich blood from the heart to the rest of the body and returns deoxygenated blood to the heart.
2. **Pulmonary circulation:** Carries deoxygenated blood from the heart to the lungs for oxygenation and brings oxygenated blood back to the heart.
3. **Coronary circulation:** Supplies blood to the heart muscle itself, ensuring it receives oxygen and nutrients.

5. Describe the coronary circuit.

The coronary circuit refers to the flow of blood in the blood vessels that supply the heart muscle (myocardium). The coronary arteries bring oxygenated blood to the heart tissue, while the coronary veins take away deoxygenated blood. This is important because the heart needs its blood supply to keep beating and pumping blood effectively.

6. Write the six functions of blood.

1. **Transportation of gases:** Blood carries oxygen from the lungs to the body and carbon dioxide back to the lungs.
2. **Transport of nutrients:** It delivers nutrients from the digestive system to all body cells.
3. **Removal of wastes:** It removes waste products like urea and transports them to the excretory organs.
4. **Transport of hormones:** Blood helps hormones travel from glands to target organs.

5. **Protection:** White blood cells in the blood fight infections and defend against disease.
6. **Clotting:** Platelets in the blood help stop bleeding by forming clots at the site of injuries.

7. What are the organs that are connected to the circulatory system?

The main organs connected to the circulatory system include the **heart**, which pumps blood; the **lungs**, where oxygen is added to the blood; the **brain**, which receives a constant blood supply; the **kidneys**, which filter blood; and the **liver**, which processes nutrients and detoxifies substances. Every part of the body is connected through blood vessels.

8. Name the types of blood vessels.

There are three main types of blood vessels:

- **Arteries:** Carry blood away from the heart and usually contain oxygen-rich blood.
- **Veins:** Carry blood back to the heart and usually contain deoxygenated blood.
- **Capillaries:** Tiny vessels where the exchange of gases, nutrients, and wastes takes place between blood and body cells.

9. What are the types of white blood cells? Why are these called white blood cells?

The main types of white blood cells are **neutrophils**, **lymphocytes**, **monocytes**, **eosinophils**, and **basophils**. They are called white blood cells because they appear white or colourless under a microscope due to the absence of haemoglobin. Their main job is to defend the body against infections and harmful invaders.

10. In what ways does the pulmonary artery differ from all other arteries?

The pulmonary artery is different because it carries **deoxygenated blood** from the heart to the lungs, while all other arteries carry **oxygenated blood** to the body tissues. It is the only artery that transports blood low in oxygen, making it a unique part of the circulatory system.

11. How are the valves of the human heart important?

Valves in the heart are important because they keep blood flowing in the **right direction**. They open to let blood move forward and close to stop it from going backwards. This ensures that the heart works efficiently and maintains a steady blood flow throughout the body.

12. Suggest why an injury that cuts open an artery is much more dangerous than an injury to a vein.

Arteries carry blood under high pressure because they come directly from the heart. If an artery is cut, blood spurts out forcefully, and blood loss happens quickly, which can be life-threatening. In contrast, veins have lower pressure, and blood loss is slower, making it less dangerous.

13. Why are valves present in veins but not in arteries?

Valves are present in veins to **prevent the backflow** of blood, especially in the lower parts of the body. Veins carry blood under low pressure and often against gravity, so valves help push the blood back toward the heart. Arteries don't need valves because the pressure from the heart pushes the blood forward.

14. Why are the walls of the atria thinner than the walls of the ventricles?

The atria have thinner walls because they only pump blood a short distance to the ventricles. This does not require much pressure. The ventricles, especially the left one, pump blood over longer distances and need thicker, stronger walls to do this.

15. Why is the muscle of the left ventricle thicker than that of the right ventricle?

The left ventricle has to pump blood to the **entire body**, which requires a lot of force, so its walls are thicker and more muscular. In contrast, the right ventricle only sends blood to the **lungs**, which are nearby, so it doesn't need as much pressure and has thinner walls.